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CSE 575: Statistical Machine Learning | K-Means Strategy Project

Project Report — K-Means Clustering (Part 1)

1. Introduction

The project uses the K-Means clustering algorithm on a given 2-D dataset and applies Strategy 1 to select an initial centroid. The algorithm is tested with $k = 2$ up to $k = 10$ clusters. The algorithm iterates until convergence to get the final centroid coordinates and the total loss when using the sum of squared Euclidean distances objective function.

2. Strategy 1: Random Initialization

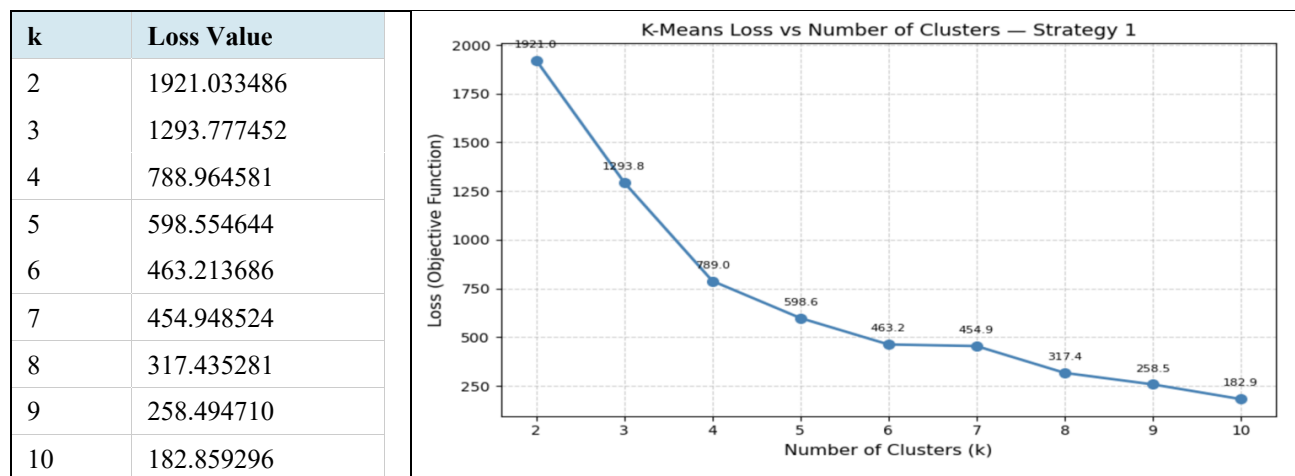
Strategy 1 uses random sampling of k data points in the dataset without replacement to pick k initial cluster centroids. The first_S1 method in precode.py is based on seeded random state that is based on the student ID (last four digits: 2599) to guarantee reproducibility. In particular, the seed is calculated as $\text{int}(\text{id}) \% 150$, and the seed of this student is 149. The algorithm then repeats between two steps (1) assigning all the points to a closest centroid using Euclidean distance, (2) and recomputing each centroid using the mean assignment of its points. This is done until convergence or 1000 repetitions.

3. Objective (Loss) Function

The loss is defined as the total sum of squared Euclidean distances from each data point to its assigned cluster centroid across all k clusters:

$$\sum_{i=1}^k \sum_{x \in D_i} ||x - \mu_i||^2$$

where μ_i is the centroid of cluster D_i and k is the number of clusters.



4. Observations

The objective function is monotonically decreasing with k (when k is varying between 2 and 10). Key observations include:

- **Steepest decline:** The largest decrease is between $k=2$ (1921.03) and $k=3$ (1293.78), the drop of about 627 units, which means that there are at least two distinct natural clusters.
- **Elbow region:** There is a significant change in rate of decrease between $k=3$ and $k=4$, indicating that $k=3$ or $k=4$ clusters might be the most effective in representing the underlying structure of the data.
- **Plateau behaviour:** The losses decrease more gradually after the $k=7$ point (317.44 to 258.49 to 182.86), showing diminishing returns as more clusters start to divide cohesive groups.

5. Analysis

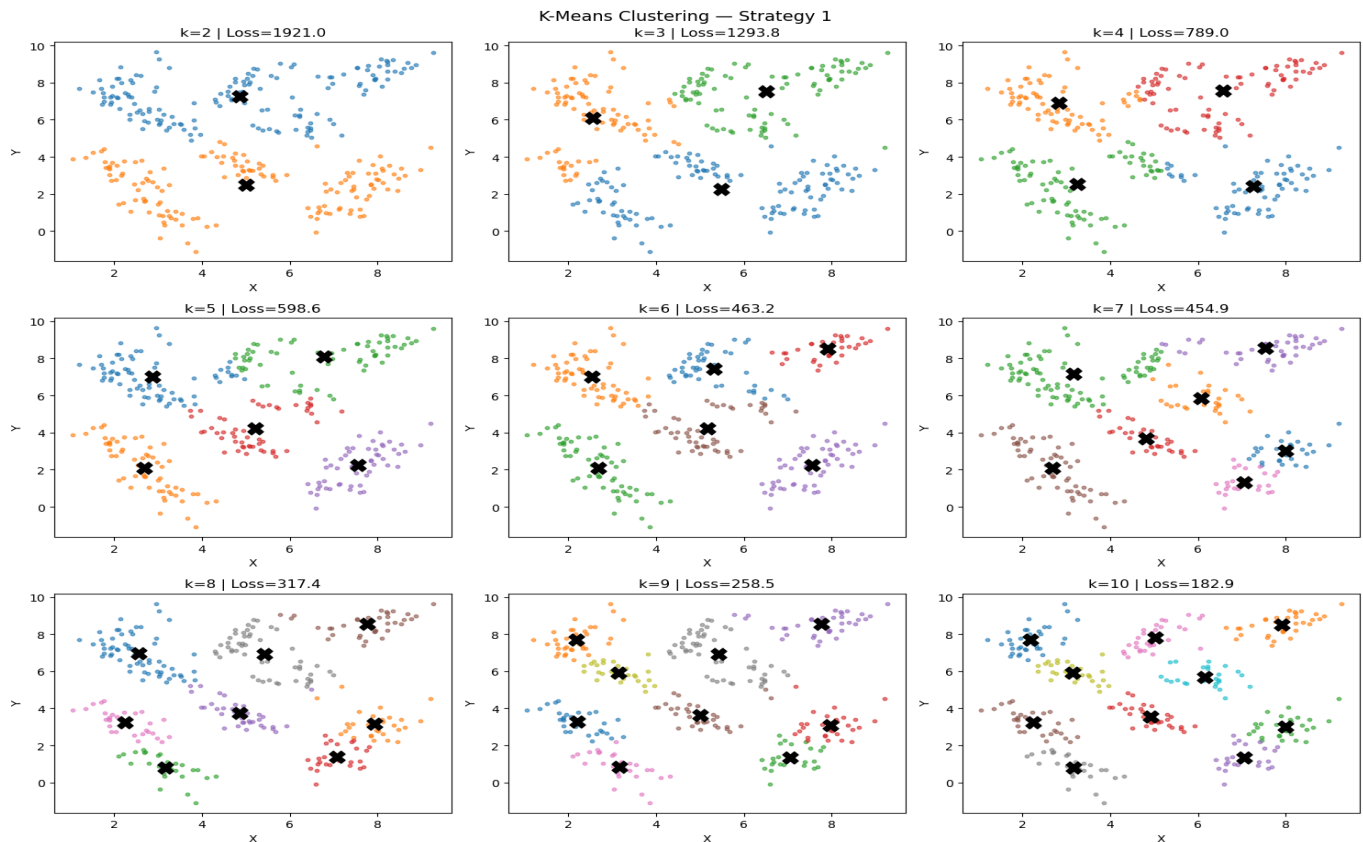
The sudden reduction in loss between $k=2$ and $k=3$ proves the idea that one pair of clusters is not sufficient to describe the structure of the dataset. The elbow of k in $[3,4]$ is a well-known heuristic to determine the best number of clusters - after this threshold, adding another cluster only splits an already well-organized cluster instead of really isolated structure.

The remarkably limited loss decrease between $k=6$ (463.21) and $k=7$ (454.95) where the loss decreased by only 8.27 is notable as an anomaly that could be explained by the fact that Strategy 1 was randomly selected. Random seeding can also be a weakness of the algorithm since, when initial centroids are placed in a sub-optimal manner, the algorithm can reach a local minimum instead of the global optimum. This is particularly so with larger values of k where the solution space is more complicated.

On the whole, Strategy 1 is good with low k values where random initialisation is less prone to degenerate solutions, but with larger k values without multiple restarts will give inconsistent results.

6. Final Centroids and Clustering Plot

The coloured X markers on the figure below show the final positions of the centroids for each value of k from 2 to 10. Each colour stands for a different cluster center. As k goes up, the centroids spread out more and more across the feature space, which is in line with the loss going down steadily.



7. Final Thoughts

This project shows that the K-Means algorithm works well for clustering the given 2D data set when it starts with random initialisation (strategy 1). The loss value goes from 1921.03 when $k=2$ to 182.86 when $k=10$. The knee on the graph is between $k=3$ and $k=4$. This means that this is the best range for representative clusters.